

NATIONAL INSTITUTE OF TECHNOLOGY ,AGARTALA



INDUSTRIAL INSTRUMENTATION LAB.

Department :Electrical Engineering

FIRE ALARM USING INFRARED

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Signature

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INTRODUCTION:

Using IR Fire Alarm is an automatic fire alarm system designed to detect the unwanted presence of fire by monitoring changes in the environment associated with flaming. The IR Receiver sensor detects the fire and generates an analog voltage. This variation is monitored by forming a potential divider circuit.

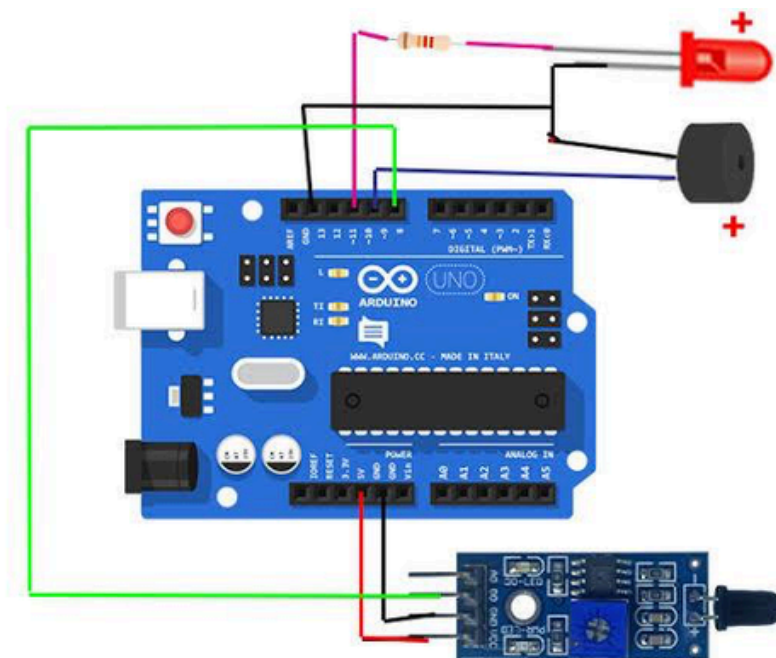
Fire Alarm Systems are very common in commercial building and factories, these devices usual contain a cluster of sensors that constantly monitors for any flame, gas or fire in the building and triggers an alarm if it detects any of these. One of the simplest way to detect fire is by using an IR Flame sensor, these sensors have an IR photodiode which is sensitive to IR light. Now, in the event of a fire, the fire will not only produce heat but will also emit IR rays, yes every burning flame will emit some level of IR light, this light is not visible to human eyes but our flame sensor can detect it and alert a microcontroller like Arduino that a fire has been detected.

In this we interface Flame Sensor with Arduino and learn all the steps to build Fire Alarm System by using Arduino and flame sensor. Flame sensor module has a photodiode to detect the light and an op-amp to control the sensitivity. It is used to detect fire and provide a HIGH signal upon the detection. Arduino reads the signal and provides alert by turning on the buzzer and LED. The flame sensor used here is an IR based flame sensor.

Components Required

S.no.	Components	Specifications	Quantity
1	Arduino	5V, ATmega328P	1
2	IR fire sensor	3.3 -5V ,Digital output	1
3	Buzzer	5V active type	1
4	LCD display	16×2, 5V	1
5	LED light	5mm , 2V	2
6	Jumper wires	M-M type	As req

Circuit Diagram



fire alarm using IR fire sensor

WORKING PRINCIPLE

The working of the fire alarm system is based on the detection of infrared (IR) radiation emitted by flames and the processing of this signal using the Arduino Uno. When combustion occurs, the flame produces infrared radiation in the wavelength range of approximately 760 nm to 1100 nm. The IR flame sensor is designed to sense this radiation and convert it into an electrical signal.

The system operates as follows:

- The flame sensor continuously monitors the surrounding environment for infrared radiation.
- Under normal conditions (no fire), the sensor output remains in its default logic state (HIGH or LOW depending on the module).
- The sensor module contains a comparator circuit that compares the detected IR intensity with a preset threshold value.

When a flame comes within the detection range, the infrared radiation falls on the sensing element (photodiode/phototransistor). This causes a change in its electrical characteristics, leading to a variation in the output voltage. Once the detected IR intensity exceeds the threshold level, the sensor output changes its logic state.

At this stage, the Arduino reads the digital signal from the sensor pin and processes it according to the programmed instructions:

- If flame is detected:
- The Arduino sends a HIGH signal to the buzzer.

- The buzzer produces an audible alarm sound.
- The LED glows to provide a visual indication.
- If no flame is detected:
 - The buzzer remains OFF.
 - The LED remains OFF.

After the flame is removed, the sensor output returns to its normal state, and the Arduino automatically turns OFF the alarm system. The process continues continuously, ensuring real-time monitoring of fire conditions.

Thus, the system converts infrared radiation into an electrical signal, processes it using a microcontroller, and generates both audio and visual alerts for safety.

Detection Modes Used in Flame Sensing

Flame detection systems generally operate in two modes: static mode and differential mode.

In static mode, the sensor detects fire based only on the intensity of infrared radiation. If the IR level crosses a predefined threshold, the system identifies it as fire. Most basic flame sensor modules used with Arduino operate in this mode. Although simple and easy to implement, static mode may sometimes produce false alarms due to sunlight or other constant IR sources.

In differential mode, the system detects rapid fluctuations in infrared radiation rather than just intensity. Real flames flicker at a frequency of approximately 5–20 Hz, producing varying IR signals. Differential mode analyzes these variations over time to confirm the presence of fire. This method reduces false alarms and improves reliability. Industrial flame detectors commonly use differential or frequency-based detection techniques.

In this project, the IR flame sensor mainly operates in static mode detection.

Arduino Code:

```
#include <Wire.h>
#include <LiquidCrystal_I2C.h>

LiquidCrystal_I2C lcd(0x27, 16, 2);

const int flamePin = 8;
const int ledPin = 6;
const int buzzerPin = 9;

int lastState = HIGH; // assume SAFE at start

void setup() {

  pinMode(flamePin, INPUT);
  pinMode(ledPin, OUTPUT);
  pinMode(buzzerPin, OUTPUT);

  digitalWrite(ledPin, LOW);
  digitalWrite(buzzerPin, LOW);

  lcd.init();
  lcd.backlight();

  lcd.setCursor(0, 0);
  lcd.print("Fire Alarm");
  lcd.setCursor(0, 1);
  lcd.print("System Ready");
  delay(2000);

  lcd.clear();
  lcd.setCursor(0, 0);
  lcd.print("Status: SAFE");
  lcd.setCursor(0, 1);
  lcd.print("No Fire Found");
}
```

```

void loop() {

    int flameState = digitalRead(flamePin);

    // ACTIVE LOW sensor (LOW = FIRE)
    if (flameState == LOW && lastState != LOW) {

        digitalWrite(ledPin, HIGH);
        digitalWrite(buzzerPin, HIGH);

        lcd.clear();
        lcd.setCursor(0, 0);
        lcd.print("!!! FIRE !!!");
        lcd.setCursor(0, 1);
        lcd.print("Take Action!");

        lastState = LOW;
    }

    else if (flameState == HIGH && lastState != HIGH) {

        digitalWrite(ledPin, LOW);
        digitalWrite(buzzerPin, LOW);

        lcd.clear();
        lcd.setCursor(0, 0);
        lcd.print("Status: SAFE");
        lcd.setCursor(0, 1);
        lcd.print("No Fire Found");

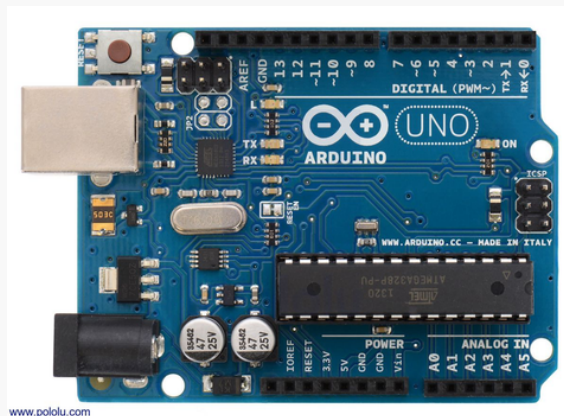
        lastState = HIGH;
    }

    delay(200);
}

```

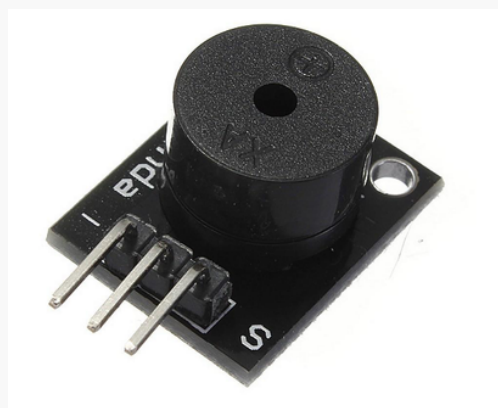
Arduino Uno

Arduino Uno is the main control unit of the fire alarm system. It is based on the ATmega328P microcontroller and operates at 5V. The board consists of 14 digital input/output pins and 6 analog input pins. In this project, the Arduino reads the digital signal from the IR flame sensor and processes it according to the program uploaded into it. When the sensor detects flame, the Arduino activates the buzzer and LED to give an alert. Thus, it acts as the brain of the entire system.



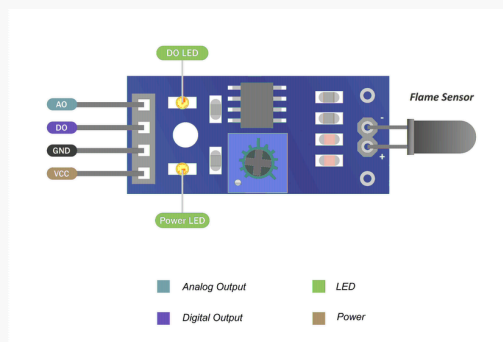
Buzzer

The buzzer is used as an audible alarm indicator. It produces sound when it receives a HIGH signal from the Arduino. In this fire alarm system, when the flame sensor detects fire, the Arduino sends a signal to the buzzer, causing it to emit a loud sound. This alerts people nearby about the presence of fire.



IR flame sensor

The IR flame sensor is used to detect fire or flame by sensing infrared radiation emitted by it. Flames emit infrared light in the wavelength range of approximately 760nm to 1100nm, which is detected by the sensor. The module typically has four pins: VCC, GND, Digital Output (DO), and Analog Output (AO). In this project, the digital output is connected to the Arduino. When flame is detected, the output signal changes state, which is then read by the Arduino to trigger the alarm system.



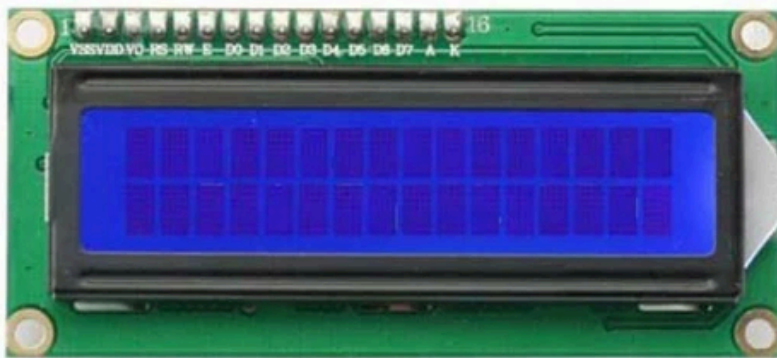
LED

An LED is used as a visual indicator in the fire alarm system. When the flame is detected, the Arduino turns on the LED along with the buzzer. The LED provides a clear visual indication of fire detection. A 220Ω resistor is connected in series with the LED to limit the current and prevent damage.



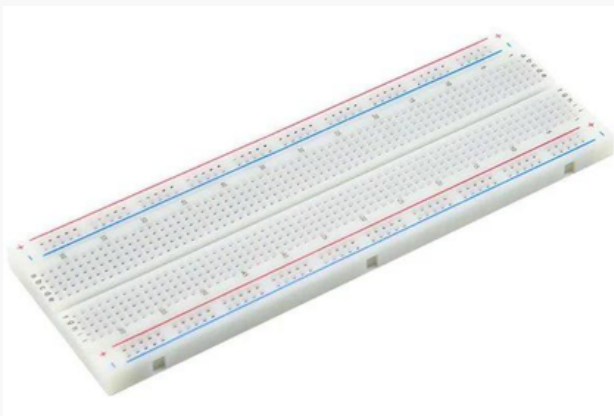
LCD Display(16×2):

A 16×2 LCD (Liquid Crystal Display) is used to show real-time information in the fire alarm system. It can display 16 characters per line across 2 lines. In this project, it is used to indicate system status such as “No Fire Detected” or “Fire Alert.” It is connected to the microcontroller (like an Arduino Uno) and helps in providing a clear visual output of sensor readings and warning messages



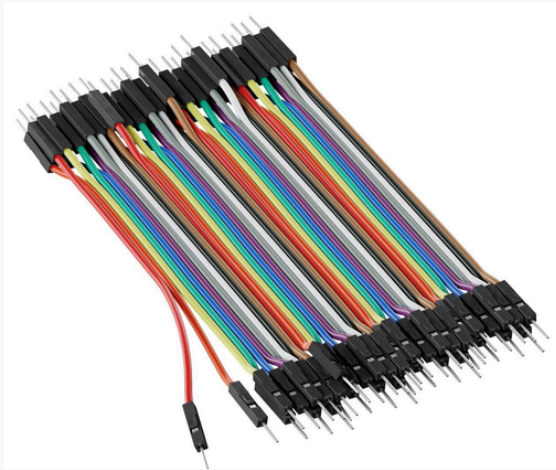
Breadboard

A breadboard is used to assemble the circuit without soldering. It allows easy placement and removal of components. All components such as the flame sensor, LED, resistor, and buzzer are connected on the breadboard using jumper wires.



Jumper wires

Jumper wires are used to establish electrical connections between Arduino, flame sensor, buzzer, LED, and breadboard. They ensure proper signal and power transmission within the circuit.



Advantages :

- Low cost
- Easy to design and implement
- Fast response time
- Suitable for small-scale safety applications

Limits:

- Sensitive to sunlight and other IR sources
- Limited detection range
- Detects flame only ,not smoke

Applications:

- Residential buildings
- Laboratories
- Industrial safety systems
- Electrical control panels

Result

The fire alarm system was successfully designed and tested. The IR flame sensor effectively detected flame within its operating range, and the Arduino Uno successfully activated the buzzer and LED as warning indicators.

Conclusion

The Fire Alarm System using IR Flame Sensor and Arduino Uno is a simple, efficient, and economical method of fire detection. The system demonstrates the practical application of infrared sensing technology and microcontroller interfacing. By implementing advanced detection techniques such as differential mode analysis, the system can be further improved for industrial-level safety applications.